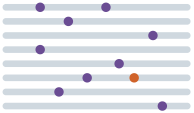
LIBRARIES
nanopore, Illumina or Sanger

AID-competent animals



AID-null controls (two or more)





nanopore Illumina Sanger

Sample sheet

Each file, or each Sanger clone, is assigned to an animal, a group, and control or not. Order and well stay with every flag.

READ PREPARATION
every read is gated

Gates

Reads longer than the amplicon, or averaging more than 0.01 errors per base, are dropped. Sanger: the longest stretch at Q ≥ 20.

Competitive alignment

Each read is aligned to the target, its paralogue and any contaminant at once, and goes to its best match. A paralogue can be analysed as a target of its own.



target paralogue other

Read accounting

The fate of every delivered read, with its reason, for every library (Figure S1).

AUTOMATED ANALYSIS
calibrated by the AID-null controls

Background

Measured in the controls, per position, change, strand and base quality, and inflated by one factor fitted leaving each control out in turn.

Candidate test

A Poisson tail over every depth of evidence, against that background. A change in half the reads or more is germline (Sanger: per sequencing batch).

Threshold set by the controls (target–decoy)

Each control is called against the others, giving decoy p-values. The threshold is the largest at which the false calls the decoys predict are at most 5% of the calls kept.

Sanger: every substitution at Q ≥ 20 counts; the AID-null clones measure the error, and each animal is told how many of its calls that background explains.

Bookkeeping checks (reported, never applied)

Copied traces and duplicate clones, as groups; clones shared between animals beyond chance; mutated control clones; label mismatches; load outliers. A trace filed under two animals on purpose is kept under both; an accidental copy is counted once.

RESULTS
per animal, then per group

Mutation profiles

unique-variant density
mutation frequency
12-type spectrum, corrected for base composition
AID (WRC/GYW) and Polη (WA/TW) hotspots
position along the amplicon
indels, clones and mutation trees
calls the background explains (Sanger)

Group tests

Welch t on $\log_{10}$ density, one value per animal, so clones of one lineage cannot stand in for independent mice.



case control

● mutation ● sequencing error